This chapter focuses on in-field reinforcement learning (Section **??**), where, unlike the simulator-based training of preceding chapters, the agent learns directly from interactions with real customers and *exploration* is a real cost that must be balanced against *exploitation*. A seller faces T customers in sequence, setting a price for each and observing only whether the customer bought.[1] The seller does not observe the customer's willingness to pay. *Regret*, the total revenue gap between the seller's policy and the best fixed price in hindsight, is the central measure: if the optimal fixed price earns r^* per customer, a policy with regret $R(T)$ earns $Tr^* - R(T)$ in total. The central question is how fast $R(T)$ shrinks as the seller accumulates purchase data, and how structural assumptions about demand affect this rate.

0.1 Foundations

0.1.1 No Structure on Demand

Kleinberg and Leighton (2003) study the simplest version of the problem. Customers arrive with valuations drawn i.i.d. from an unknown distribution on $[0, 1]$, and the seller posts a price from a continuous set. The demand curve $D(p) = \Pr(v \geq p)$ is unknown; the only feedback is whether each customer bought. The seller's goal is to minimize regret against the single price p^* that maximizes $p \cdot D(p)$. Kleinberg and Leighton prove that the minimax regret is $\Theta(\sqrt{T})$.[2] In concrete terms, after 10,000 customers the seller loses roughly 100 customers' worth of revenue to the uncertainty in demand. No algorithm can do better without imposing structure on D. For adversarial valuations (chosen by a worst-case opponent rather than drawn from a fixed distribution), the minimax regret rises to $\Theta(T^{2/3})$, achieved by the Exp3 algorithm[3] on a discretized price grid. I focus on the stochastic setting throughout this chapter.

[1] Rothschild (1974) posed pricing under demand uncertainty as a two-armed bandit problem. His insight was that a *myopic* seller can get stuck at a suboptimal price forever, because exploiting the currently best-looking price generates no information about alternatives.

[2] The upper bound, $O(\sqrt{T \log T})$, discretizes $[0, 1]$ into $K = \lceil (T/\log T)^{1/4} \rceil$ prices and runs the UCB1 algorithm of Auer et al. (2002). The lower bound, $\Omega(\sqrt{T})$, constructs a family of demand curves parameterized by the location of the optimal price $p^* \in [0.3, 0.4]$. The key tension: posting prices far from p^* is informative about demand but costly in revenue; posting prices near p^* is cheap but uninformative. Resolving this tension costs at least $\Omega(\sqrt{T})$ in cumulative revenue. UCB1 (Auer et al., 2002) selects the price maximizing $\hat{\mu}_{p_k}(t) + \sqrt{2 \ln t / N_{p_k}(t)}$, where $\hat{\mu}_{p_k}(t)$ is the empirical mean profit and $N_{p_k}(t)$ the number of trials; the second term is an exploration bonus that shrinks as a price is tried more, implementing the principle of optimism in the face of uncertainty.

[3] *Exp3* (**?**) maintains a weight $w_k(t)$ for each price, selecting p_k with probability proportional to $w_k(t)$ and updating the chosen price's weight by $\exp(\eta \hat{r}_{k,t})$ where $\hat{r}_{k,t}$ is the revenue importance-weighted by the selection probability; because no model of demand is assumed, the guarantee holds against an adversary who chooses valuations after observing the algorithm.

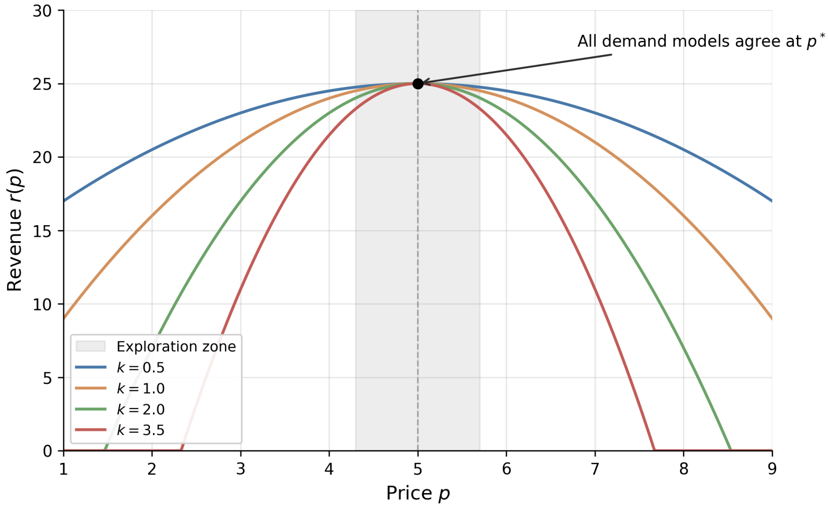

Figure 1: Revenue curves $r(p) = r^* - k(p - p^*)^2$ for four demand curvatures $k \in \{0.5, 1.0, 2.0, 3.5\}$. All models agree at the optimal price $p^* = 5$. Within the shaded exploration zone, the curves are nearly indistinguishable, so playing prices near p^* is uninformative about the demand parameter.

0.1.2 Parametric Demand

Broder and Rusmevichientong (2012) consider a parametric demand model $d(p; z) = \Pr(V \geq p)$, where $z \in \mathbb{R}^n$ is an unknown parameter governing the demand curve. The seller observes binary purchase decisions and updates a maximum likelihood estimate of z. Under standard regularity conditions (bounded demand, unique optimal price, smooth revenue function), the minimax regret is again $\Theta(\sqrt{T})$.[4] Parametric structure alone does not break the $\sqrt{T}$ barrier.

The picture changes under a "well-separated" condition requiring that every price in the feasible set is informative about the demand parameter. Formally, the Fisher information $I(p, z)$ is bounded below by $c_f > 0$ for all prices p and parameters z.[5] Under well-separation, a pure greedy policy (MLE-Greedy, pricing at $p^*(\hat{z}_t)$ each period) achieves $O(\log T)$ regret (Theorem 4.8), because every price is informative and dedicated exploration rounds become unnecessary.

[4] The lower bound (Theorem 3.1 of Broder and Rusmevichientong (2012)) constructs a linear demand family where all demand curves pass through the same point at the optimal price $p^*(z_0)$. Observing purchases at this price provides no information about z. An MLE-Cycle policy that interleaves dedicated exploration rounds with greedy pricing achieves the matching upper bound $O(\sqrt{T})$ (Theorem 3.6).

[5] This means no two demand curves $d(p; z)$ and $d(p; z')$ cross on the pricing interval, so every purchase observation distinguishes the two hypotheses. In the lower-bound family of Theorem 3.1, the curves all cross at $p = 1$, which is why the $\sqrt{T}$ floor emerges.

0.1.3 High-Dimensional Features with Sparsity

Javanmard and Nazerzadeh (2019) extend the setting to products described by d-dimensional feature vectors x_t. The market value is $v_t = x_t^\top \theta_0 + z_t$, where $\theta_0 \in \mathbb{R}^d$ is unknown and z_t is i.i.d. noise from a known log-concave distribution F.[6] Only s_0 of the d coordinates of θ_0 are nonzero, but the seller does not know which ones.

Their RMLP algorithm (Regularized Maximum Likelihood Pricing) operates in episodes whose lengths double (1, 2, 4, 8, . . .). At each episode boundary, the seller fits a LASSO-penalized maximum likelihood estimate of θ_0 using data from the previous episode, then prices greedily throughout the current episode. The regret is $O(s_0 \log d \cdot \log T)$ (Theorem 4.1), with a matching lower bound of $\Omega(s_0(\log d + \log T))$ (Theorem 5.1). The reason this beats $\sqrt{T}$ connects back to the Broder lower bound. In the parametric model of Section 0.1.2, all demand curves can cross at the optimal price, making that price uninformative. Here, customer features vary across periods, so the aggregate demand function at any fixed price changes in proportion to the estimation error in θ_0. Every price is informative, and dedicated exploration rounds become unnecessary.[7] Even with hundreds of features, if only a handful matter, learning is fast.

0.2 Revealed Preference and Partial Identification

Misra et al. (2019) bring economic theory into the bandit pricing framework. Their model has K discrete prices, S consumer segments (segment membership observed, but valuations unknown), and within-segment heterogeneity δ. A consumer in segment s has valuation $v_i = v_s + n_i$, where v_s is the segment midpoint and $n_i \in [-\delta, \delta]$ is idiosyncratic noise. The key structural assumption is WARP (Weak Axiom of Revealed Preference): if a consumer buys at price p, she would buy at any lower price $p' < p$.

WARP enables partial identification of each segment's valuation. Suppose the seller has offered several prices to segment s. Define $p_s^{\min} = \max\{p_k : \text{all customers purchased}\}$ and $p_s^{\max} = \min\{p_k : \text{no customer purchased}\}$. Then the segment midpoint lies in

[6] Log-concavity of F and $1-F$ is satisfied by the normal, logistic, uniform, Laplace, and exponential distributions. It ensures that expected revenue $p \cdot [1 - F(p - x_t^\top \theta_0)]$ is strictly quasi-concave in p, giving a unique optimal price.

[7] If some feature directions are rarely observed, the seller cannot learn all coordinates of θ_0 quickly, and regret degrades to $O(\sqrt{\log(d) \cdot T})$ (Theorem 4.2). If the noise distribution belongs to a known parametric family but its scale parameter is unknown, regret reverts to $\Omega(\sqrt{T})$ (Theorem 7.1), foreshadowing the result of Xu and Wang (2021) discussed in Section 0.3.

$[p_s^{\min}, p_s^{\max}]$, and within-segment heterogeneity satisfies $\hat{\delta}_s \leq (p_s^{\max} - p_s^{\min})/2$. These bounds propagate to aggregate demand: for each price p_k, the seller constructs upper and lower bounds on total profit $\pi(p_k) = p_k \cdot D(p_k)$. When the profit upper bound at some price falls below the best profit lower bound across all prices, that price is dominated and permanently eliminated from consideration.

The UCB-PI algorithm combines dominance elimination with a price-scaled exploration bonus:

$$I_{p_k}(t) = \hat{\mu}_{p_k}(t) + p_k \sqrt{\frac{2 \ln t}{N_{p_k}(t)}} \tag{1}$$

if p_k is not dominated, and $I_{p_k}(t) = 0$ otherwise, where $\hat{\mu}_{p_k}(t)$ is the average profit observed at price p_k and $N_{p_k}(t)$ is the number of trials.[8] Two innovations drive the improvement over standard UCB1. First, dominance elimination reduces the effective number of arms the algorithm must explore. Second, the p_k scaling focuses exploration on prices where uncertainty actually matters for profit. Together, these yield $O(\log T)$ regret.

$$\mathbb{E}[R_T(\text{UCB-PI})] \leq \sum_{k \neq k^*} \frac{8 p_k \log T}{\Delta_k} + \left(1 + \frac{\pi^2}{3}\right) \sum_{k=1}^{K} \Delta_k \tag{2}$$

where $\Delta_k = \mu_{k^*} - \mu_k$ is the gap between arm k and the optimal arm.[9]

Misra et al. (2019) calibrate the model to an in-field deployment at ZipRecruiter, a B2B online recruiting platform. With 7,870 customers per month, 1,000 segments, and 10 price points from \$19 to \$399, UCB-PI achieves 98% of oracle profit and produces 43% higher profits during the first month of testing compared to a learn-then-earn alternative. The algorithm has both higher mean profit and lower variance, reflecting the value of eliminating dominated prices early.[10]

0.3 The Value of Knowing the Noise Distribution

Xu and Wang (2021) ask how much it helps to know the shape of demand uncertainty.

[8] Standard UCB1 uses an exploration bonus of $\sqrt{2 \ln t / N_{p_k}(t)}$, which assumes rewards in $[0, 1]$. Since profit at price p_k is bounded by p_k, scaling the bonus by p_k tightens exploration for cheap prices that cannot contribute much profit regardless.

[9] The first sum is the leading term, scaling as $O(\log T)$. The second sum is a constant that does not grow with T. Replacing p_k with 1 recovers the standard UCB1 bound, which is looser.

[10] For multi-product settings, Mueller et al. (2019) impose low-rank structure on the price-sensitivity matrix, achieving regret $O(T^{3/4}\sqrt{d})$ that scales with the latent demand dimension d rather than the number of products. Badanidiyuru et al. (2013) extend the bandit framework to handle inventory constraints ("bandits with knapsacks"), relevant when the seller faces limited stock alongside the pricing decision.

In their model, a feature vector $x_t \in \mathbb{R}^d$ describes each sales session, the customer's valuation is $w_t = x_t^\top \theta^* + N_t$ where θ^* is unknown and N_t is zero-mean i.i.d. noise with CDF F, and the seller observes only whether the customer bought at the posted price. The answer depends on whether F is known.[11]

If F is known (the seller knows that demand shocks are, say, normally distributed with known variance), the EMLP algorithm (Epoch-based Maximum Likelihood Pricing) achieves regret $O(d \log T)$ (Theorem 3).[12] After 10,000 customers with $d = 5$ features, the revenue loss is roughly 50 customers' worth, an improvement over the $\sqrt{T} \approx 100$ customers' worth that Kleinberg's lower bound imposes without structural knowledge.

If F is unknown (even if only the variance of a Gaussian is unknown, with everything else known), the regret is at least $\Omega(\sqrt{T})$ (Theorem 12).[13] The seller is back to the Kleinberg baseline, with no algorithm able to achieve sublinear improvement regardless of how many features are available.

The gap between $O(d \log T)$ and $\Omega(\sqrt{T})$ is super-polynomial in T, not merely a constant factor. When a modeler specifies a logit or probit demand model, the assumed noise distribution purchases a qualitative improvement in learning rate. Semiparametric approaches that leave the error distribution unspecified pay a concrete cost, reverting from logarithmic to polynomial regret.[14]

0.4 Strategic Buyers

Liu et al. (2024) introduce buyer manipulation into contextual dynamic pricing. At time t, a buyer arrives with true covariates $x_t^0 \in \mathbb{R}^d$ and valuation $v_t = \theta_0^\top x_t^0 + z_t$. The

[11] The regret benchmark here differs from Section 0.1.1. Kleinberg and Leighton (2003) and Broder and Rusmevichientong (2012) measure regret against the best fixed price; Xu and Wang (2021) and Liu et al. (2024) measure regret against the clairvoyant contextual policy that sets the optimal price p_t^* for each customer's features x_t. The contextual benchmark is harder, which makes the $O(d \log T)$ rate more impressive.

[12] EMLP runs in doubling epochs of length $\tau_k = 2^{k-1}$. At each epoch boundary, the seller fits a maximum likelihood estimate $\hat{\theta}_k$ using data from the previous epoch, then prices greedily at $p_t = J(x_t^\top \hat{\theta}_k)$ throughout the epoch, where $J(u) = \arg\max_v \, v[1 - F(v - u)]$ is the revenue-maximizing price function. The key technical insight is that the negative log-likelihood is strongly convex (Lemma 7 of Xu and Wang (2021)), so MLE concentrates at rate $O(d/\tau_k)$. Since regret is quadratic in the parameter estimation error (Lemma 5) and there are $O(\log T)$ epochs, the total regret is $O(d \log T)$.

[13] The lower bound constructs two noise variances $\sigma_1 = 1$ and $\sigma_2 = 1 - T^{-1/4}$. Any algorithm that performs well under both must spend $\Omega(\sqrt{T})$ revenue distinguishing the two cases. This extends the "uninformative price" phenomenon of Broder and Rusmevichientong (2012): when the seller does not know F, there exist prices at which observed purchase behavior is nearly identical under different demand parameters.

[14] Tullii et al. (2024) establish the tightest known bound under minimal distributional assumptions: if the noise distribution (c.d.f.) is merely Lipschitz continuous, the minimax regret is $\Theta(T^{2/3})$, strictly between the $\log T$ rate with known F and the $\sqrt{T}$ rate with unknown F. Fan et al. (2024) consider a semiparametric setting where the noise density is smooth and connect the pricing problem to the econometrics of semiparametric estimation.

seller announces a pricing rule $p_t = g(\hat{\theta}_k^\top x_t)$, where $\hat{\theta}_k$ is the current parameter estimate. Crucially, the buyer observes this rule and can distort her reported features. She solves a cost-minimization problem:

$$\min_{\tilde{x}} \, (p - v_t) + \frac{1}{2}(\tilde{x} - x_t^0)^\top A(\tilde{x} - x_t^0) \tag{3}$$

where A is a positive definite matrix governing manipulation costs.[15] The first-order condition yields a systematic bias: the buyer shifts her features to make the seller's model predict a lower valuation, securing a lower price. The seller observes only the distorted features $\tilde{x}_t$, not the true x_t^0.

Theorem 1 (Theorem 1 of Liu et al. (2024))**.** *Under standard regularity conditions, any pricing policy that treats reported features as truthful accumulates regret $\Omega(T)$.*

The regret is linear in T: every standard dynamic pricing algorithm, including EMLP and RMLP, systematically underprices because it bases decisions on manipulated features, and the bias does not shrink with more data because the manipulation is endogenous to the pricing rule.

The fix is to jointly estimate demand parameters and manipulation behavior. Liu et al. (2024) propose an episodic algorithm with two phases per episode. During the exploration phase, the seller posts uniform random prices that do not depend on features. Since the price is independent of $\tilde{x}_t$, buyers have no incentive to manipulate, and the seller observes true features x_t^0. During the exploitation phase, the seller uses the corrected pricing rule that anticipates the manipulation:

$$p_t = g\left(\hat{\theta}_k^\top x_t + \hat{\beta}_k^\top A^{-1}\hat{\beta}_k \cdot g'(\hat{\theta}_k^\top x_t)\right) \tag{4}$$

where $\hat{\beta}_k$ is the estimated coefficient on the manipulable features and g is the optimal pricing function. This correction adds a markup that offsets the anticipated feature distortion.

Theorem 2 (Theorem 2 of Liu et al. (2024))**.** *With known manipulation cost matrix A, the strategic pricing algorithm achieves regret $O(d\sqrt{T})$.*

[15] The matrix A captures how costly it is for the buyer to distort each feature dimension. High eigenvalues of A mean manipulation is expensive. This is the standard model of strategic classification (Hardt et al., 2016), adapted to pricing.

When A is unknown, the seller can still recover $O(d\sqrt{T/\tau})$ regret by tracking repeat buyers across exploration and exploitation phases, where τ is the fraction of buyers who appear in both phases (Theorem 3). Higher repeat rates mean more matched pairs for estimating manipulation behavior.

Incentive compatibility matters even in settings where the seller is "just" learning demand.[16]

0.5 Comparison of Regret Rates

Table 1 collects regret rates ordered from weakest to strongest structural assumptions. The dominant pattern is that stronger assumptions yield faster learning, with the gap between $\log T$ and $\sqrt{T}$ being super-polynomial in T, not merely a constant factor. Strategic behavior is the outlier: it produces linear regret that no amount of data can overcome without explicit correction. Figure 2 plots each rate on a log-log scale.

Table 1: Regret rates in dynamic pricing under progressively stronger assumptions. T is the number of customers, d the feature dimension, s_0 the sparsity level. The last column translates asymptotic rates into concrete terms for $T = 10{,}000$ with $d = 5$, setting constants to 1.

Paper	Demand	Noise	Regret	Per 10K (d=5)
Kleinberg and Leighton (2003)	none	none	$\sqrt{T}$	~100 lost
Broder and Rusmevichientong (2012)	parametric	known family	$\sqrt{T}$	~100 lost
Broder and Rusmevichientong (2012)	param., well-sep.	known family	$\log T$	~9 lost
Javanmard and Nazerzadeh (2019)	linear, s_0-sparse	known, log-conc.	$s_0 \log d \log T$	fast if s_0 small
Xu and Wang (2021)	linear, contextual	known	$d \log T$	~46 lost
Xu and Wang (2021)	linear, contextual	unknown var.	$\geq \sqrt{T}$	~100 lost
Tullii et al. (2024)	linear, contextual	Lipschitz only	$T^{2/3}$	~464 lost
Misra et al. (2019)	WARP	–	$\log T$	~9 lost
Liu et al. (2024)	linear + strategic	known, naïve	T	never improves
Liu et al. (2024)	linear + strategic	known, corrected	$d\sqrt{T}$	~500 lost

0.6 Applications

0.6.1 Joint Assortment and Pricing at Scale

Cai et al. (2023) tackle the joint assortment-pricing problem, where a retailer must simultaneously choose which products to display and at what prices. With a large catalog

[16] Agrawal and Tang (2024) document a related phenomenon in pricing with reference effects. If consumers anchor on past prices, a static pricing policy that ignores reference dependence accumulates linear regret $\Omega(T)$. Chen et al. (2025) show that imposing fairness constraints (requiring similar prices for similar customers) raises the regret floor to $\Theta(T^{2/3})$, a social cost of equitable treatment.

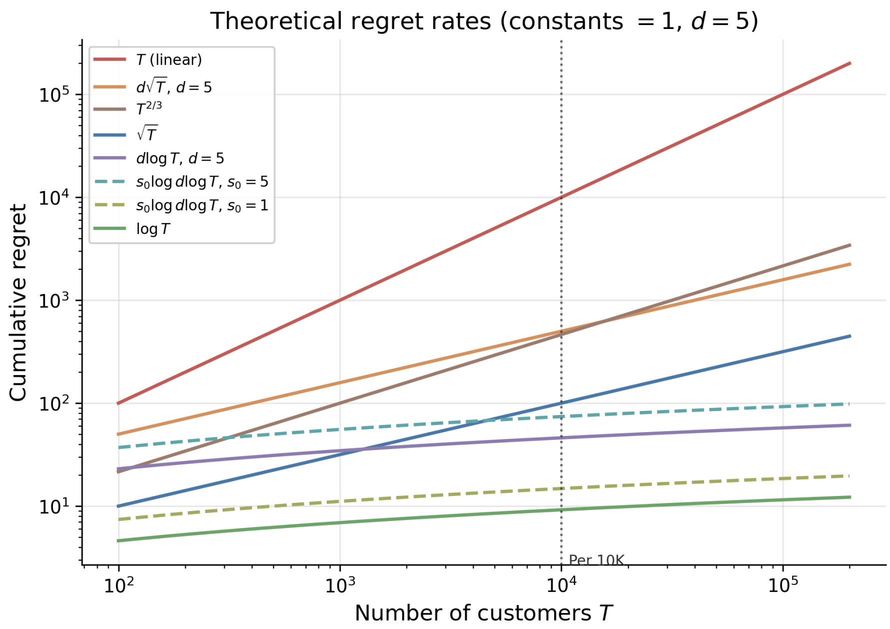

Figure 2: Theoretical regret rate functions at constants equal to 1, $d = 5$. Two cases for the $s_0 \log d \log T$ rate are shown: $s_0 = 1$ (very sparse, ≈ 15 lost at $T = 10{,}000$) and $s_0 = 5$ (moderate sparsity, ≈ 74 lost). The vertical line marks $T = 10{,}000$, matching the "Per 10K" column of Table 1.

and limited shelf space, the number of possible assortments is combinatorially vast; for a Chinese instant noodle producer with 176 products and 30 display slots, there are $\binom{176}{30} \approx 6.4 \times 10^{33}$ possible assortments before prices are even set. Customer demand also depends on market context such as region and season. In each period t, the retailer selects an assortment-pricing vector $a_t \in \mathbb{R}^{d_a}$ (encoding which products to display and at what prices), observes a context vector $x_t \in \mathbb{R}^{d_x}$ (encoding customer demographics and market conditions), and earns revenue Y_t. Cai et al. model expected revenue as $\mathbb{E}[Y_t \mid x_t, a_t] = a_t^\top \Theta^* x_t$, where $\Theta^* \in \mathbb{R}^{d_a \times d_x}$ is an unknown matrix assumed to have low rank $r \ll \min\{d_a, d_x\}$. The low-rank assumption is the demand-side analogue of factor models in asset pricing; a small number of latent dimensions (flavor preferences, seasonal effects, price sensitivity) explain most of the variation in purchasing behavior.

Their Hi-CCAB algorithm estimates Θ^* via penalized least squares, where the penalty is the nuclear norm (the sum of singular values) of Θ. This penalty encourages low-rank solutions, playing the same role for matrices that the LASSO penalty plays for sparse vectors. The time-averaged regret is $\tilde{O}(T^{-1/6})$ with dimension dependence $r(d_a + d_x)$ rather than $d_a \cdot d_x$, so the effective parameter count scales with the number of latent factors, not the full product-by-context matrix. In simulation, Hi-CCAB achieves nearly four times the cumulative sales of the noodle producer's historical assortment strategies, averaged over 100 replications.

Ganti et al. (2018) deploy Thompson Sampling in-field for dynamic pricing at Walmart.com. Their MAX-REV-TS algorithm models demand for each item i on day t via a constant-elasticity function $d_{i,t}(p) = f_{i,t}(p/p_{i,t-1})^{\gamma_i^*}$, where $d_{i,t}(p)$ is unit sales at price p, $f_{i,t}$ is a baseline demand forecast at the previous day's price $p_{i,t-1}$, and $\gamma_i^* < -1$ is the unknown price elasticity. The structural assumption, that demand responds to price through a single elasticity parameter per item, reduces the learning problem from estimating a full demand curve to estimating one scalar per item. MAX-REV-TS places a Gaussian prior over the elasticity vector γ^* and draws posterior samples at each period to solve a constrained revenue maximization problem. In a five-week in-field experiment on a basket of roughly 5,000 items, Thompson Sampling produced a statistically significant increase in per-item revenue relative to the passive pricing baseline.

0.7 Simulation Study: The Knowledge Ladder

I run six algorithms on the Misra et al. (2019) demand environment to trace how cumulative regret responds to increasing structural knowledge.[17] The six algorithms, ordered by the structural knowledge they exploit, are: (0) ε-greedy with $\varepsilon = 0.1$, which never adapts its exploration rate; (1) Learn-Then-Earn (LTE), which explores uniformly for the first 5% of rounds and then commits to the empirical best price; (2) UCB1 (Auer et al., 2002), which adapts exploration via confidence bounds but ignores demand structure; (3) Thompson Sampling (Thompson, 1933), which maintains a Bayesian posterior over purchase rates[18]; (4) UCB-PI (Misra et al., 2019), which uses WARP to eliminate dominated prices and scales the exploration bonus by the price level; and (5) UCB-PI-tuned, which adds a variance-based refinement to the exploration bonus.

Figure 3 reports cumulative regret at four checkpoints. The results confirm the theoretical progression from Table 1: ε-greedy grows linearly, UCB1 and Thompson Sam-

[17] The environment has $K = 100$ prices on a grid from \$0.01 to \$1.00, $S = 1{,}000$ consumer segments with equal weights, within-segment heterogeneity $\delta = 0.1$, and segment midpoints $v_s \sim \text{Uniform}(0.1, 0.9)$. A consumer purchases if and only if $v_i \geq p$ (WARP). Each algorithm runs across 10 seeds with $T = 200{,}000$ rounds.

[18] At each period the algorithm draws one sample $\tilde{\mu}_k$ from each arm's posterior over its purchase rate and selects the arm with the highest sampled expected profit $p_k\tilde{\mu}_k$; arms with uncertain posteriors have high-variance draws and are selected frequently, while well-estimated arms are selected in proportion to how likely they are optimal.

pling grow as $\sqrt{T}$, and UCB-PI-tuned achieves the lowest regret at every checkpoint past $T = 10{,}000$. The untuned UCB-PI variant overexplores, scaling as $\sqrt{T}$ rather than the theoretical $\log T$, illustrating that WARP-based elimination alone is insufficient without variance-calibrated exploration bonuses.

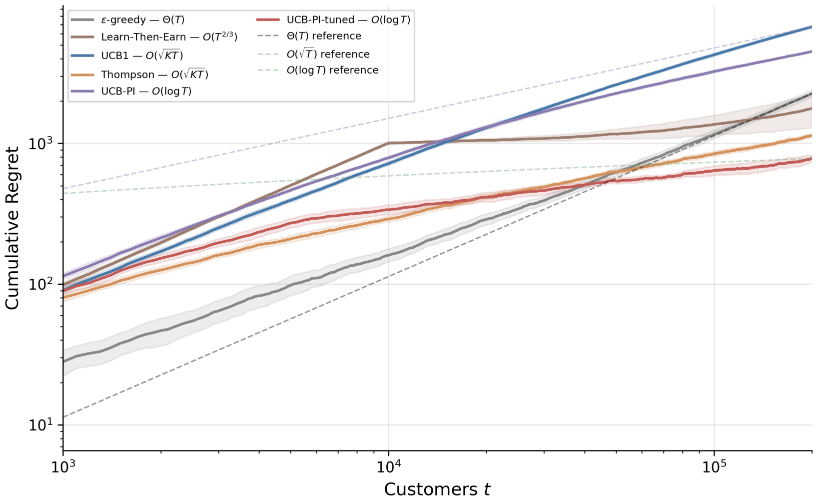

Figure 3: Cumulative regret on log-log axes ($K = 100$, $S = 1{,}000$, 10 seeds). Each algorithm's legend entry includes its theoretical regret rate. Dashed lines show $\Theta(T)$, $O(\sqrt{T})$, and $O(\log T)$ reference rates. Shaded regions are ± 2 standard errors.

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